Geochemistry Geophysics Geosystems G3

Published by AGU and the Geochemical Society

Article
Volume 14, Number 7
19 July 2013
doi: 10.1002/ggge.20142
ISSN: 1525-2027

Paleointensity results from the late-Archaean Modipe Gabbro of Botswana

A. R. Muxworthy
Department of Earth Science and Engineering, Imperial College London, London, UK (adrian.muxworthy@imperial.ac.uk)

M. E. Evans
Institute for Geophysical Research, University of Alberta, Edmonton, Alberta, Canada

S. J. Scourfield
Department of Earth Science and Engineering, Imperial College London, London, UK

J. G. King
Department of Physics, University of Botswana, Gaborone, Botswana

[1] Rock-magnetic and Thellier-Thellier-Coe paleointensity results are reported for a new collection of samples from the 2.78 Ga Modipe Gabbro of Botswana. The magnetic properties are very favorable, leading to an unusually high success rate and a well-constrained result of 36–40 μT (95% confidence interval). We discuss the long-standing problem of allowing for the enormous differences between the natural and laboratory cooling rates, and apply a temperature-dependent correction derived from first-order reversal curve (FORC) data. Whereas pure single-domain (SD) corrections often lead to quite large decreases in paleointensity estimates (sometimes exceeding 50%), we find a modest increase of about 10%. The Earth's Magnetic Dipole moment in the late Archaean is thereby estimated to have been $\sim 6 \times 10^{22}$ Am^2.

Components: 4,098 words, 8 figures, 3 tables.

Keywords: Modipe Gabbro; paleointensity; late-Archaean.

Index Terms: 1521 Geomagnetism and Paleomagnetism: Paleointensity; 1507 Geomagnetism and Paleomagnetism: Core processes; 1540 Geomagnetism and Paleomagnetism: Rock and mineral magnetism.

Received 28 January 2013; **Revised** 1 April 2013; **Accepted** 9 April 2013; **Published** 19 July 2013.

Muxworthy, A. R., M. E. Evans, S. J. Scourfield, and J. G. King (2013), Paleointensity results from the late-Archaean Modipe Gabbro of Botswana, *Geochem. Geophys. Geosyst., 14,* 2198–2205, doi:10.1002/ggge.20142.

1. Introduction

[2] In a recent paper [*Muxworthy and Evans*, 2013], we describe how micromagnetic calculations and rock-magnetic measurements on mineral separates combine to indicate that the Modipe Gabbro, an Archaean intrusion in southern Africa, is an ideal magnetic recorder likely to yield a robust estimate of the planetary magnetic dipole moment at an early stage of Earth history. This, in turn, will help constrain models of the thermal and geodynamic evolution of the Earth, its internal structure, and even the development of a habitable planetary surface [*Biggin et al.*, 2009; *Tarduno*, 2009; *Aubert et al.*, 2010; *Tarduno et al.*, 2010]. Furthermore, the value of a successful paleointensity estimate

© 2013. American Geophysical Union. All Rights Reserved.

will be enhanced by the fact that the age of this intrusion has now been very accurately determined to be 2784.0 ± 1.5 Ma [*Feinberg et al*., 2010]. Here we report a series of paleointensity and related rock-magnetic experiments obtained from an entirely new collection of samples.

2. Methods

[3] In October 2012, we collected 39 oriented drill cores and a single unoriented ~2 kg hand block from the same site in Botswana (Figure 1) investigated by *Evans and McElhinny* [1966] and *Evans et al*. [1968]. These latter authors described the rock as a medium-grained Gabbro consisting primarily of euhedral to subhedral plagioclase and anhedral pyroxene. They also demonstrated that the dominant carrier of stable magnetic remanence is magnetite in the form of tiny pyroxene-hosted inclusions. These conclusions were confirmed by modern analysis on the samples collected by *Evans and McElhinny* [1966] involving FORC measurements and micromagnetic simulations [*Muxworthy and Evans*, 2013]. For the present investigation, cylindrical specimens of two different sizes were used: standard paleomagnetic specimens (25.4 mm diameter, 25.4 mm length) and mini-cores (11 mm diameter, 10 mm length). For directional analysis, alternating-field (AF) and thermal (TH) demagnetization experiments were carried out with a static DTECH AF demagnetizer and an ASC TD48 paleomagnetic oven, respectively. Remanences were measured with an Agico JR5A spinner magnetometer. Hysteresis parameters (coercive force B_c, coercivity of remanence B_{cr}, saturation magnetization M_s, saturation remanence M_{rs}), first-order reversal curve (FORC) distributions, and thermomagnetic curves were obtained with a Princeton Measurements vibrating sample magnetometer (VSM) fitted with a furnace. Paleointensity experiments were carried out in an Orion three-axis low-field VSM using the double-heating Thellier-Thellier-Coe protocol with partial thermoremanent magnetization (pTRM) and pTRM-tail checks, using a laboratory field of 40 A/m (50.3 μT).

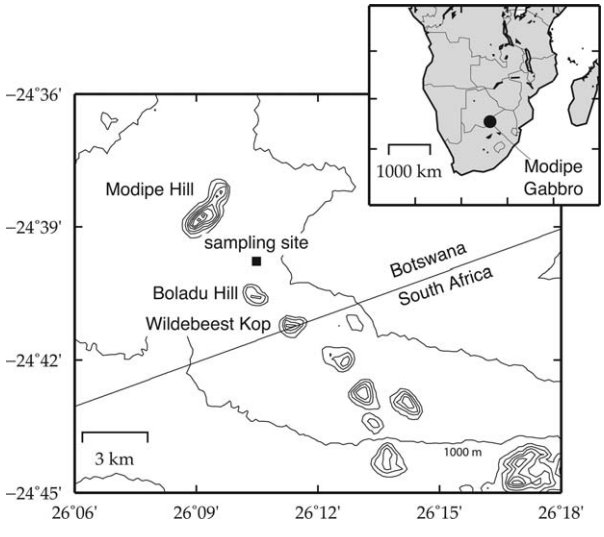


Figure 1. Sampling site location (24°39'481S, 26°10'291E). The contour spacing is 40 m, and the 1000 m contour is marked.

3. Results

3.1. Directional and Rock-Magnetic Experiments

[4] Five standard paleomagnetic specimens were progressively demagnetized (two AF, three TH) to check the stability of the Natural Remanent Magnetization (NRM). Figure 2 shows an AF example. The other four gave similar directions, and the overall mean is D=192°, I=70°, α_{95}= 5° and k = 217.

[5] Figure 3 shows hysteresis ratios on a "Day" plot [*Day et al*., 1977]. For the most part, the data define a trend in the pseudosingle domain (PSD)

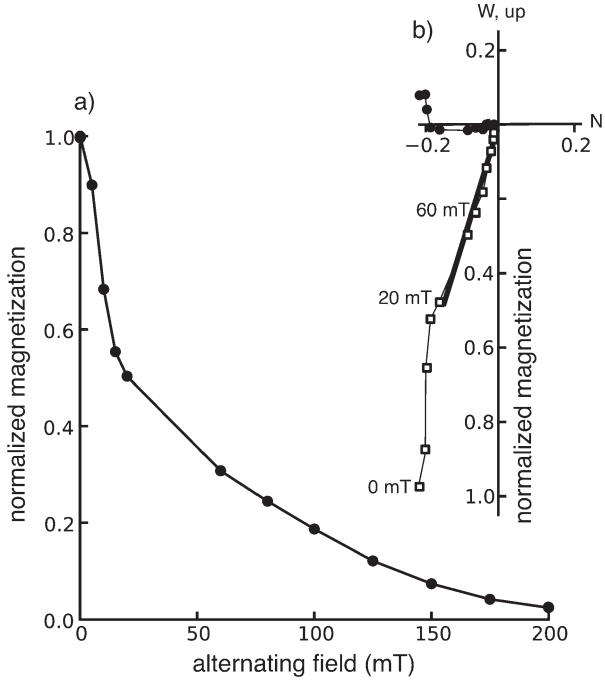


Figure 2. Example of alternating-field demagnetization. The initial natural remanent magnetization (NRM) intensity is 1.94 A/m. The mean destructive field (MDF) for the sample is ~20 mT.

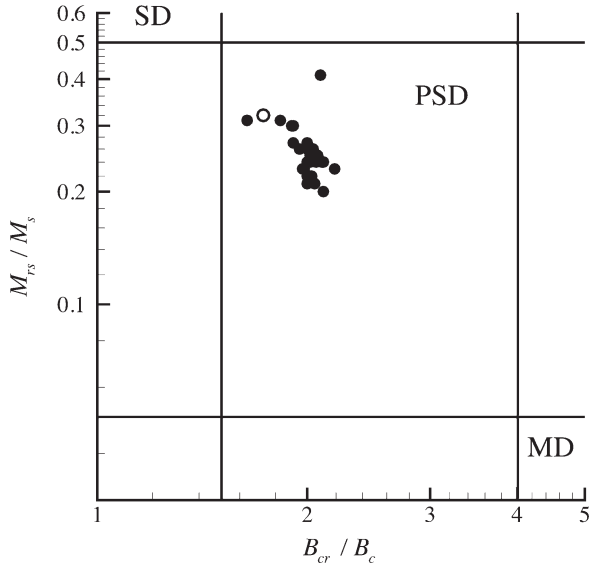


Figure 3. Hysteresis ratio 1Day1 plot for 31 specimens from the new collection (black dots), and for the hand-picked pyroxene grains (sample A4HP) used in our earlier paper [*Muxworthy and Evans*, 2013] (open circle).

field, which mimics that found in the pyroxene mineral separates previously studied (see Figure 3 in *Muxworthy and Evans* [2013]). For comparison, we include the end-member of that sequence containing the "purest" (hand-picked) pyroxene grains. It is well known that hysteresis ratio plots are nonunique; points in the PSD field can always be attributed to suitable SD-MD mixtures. Indeed, occasional large magnetite grains are present in the Modipe Gabbro [*Evans et al.*, 1968]. But the ultimate success of the paleointensity experiments described below strongly suggests that they must play a very minor role as stable remanence carriers.

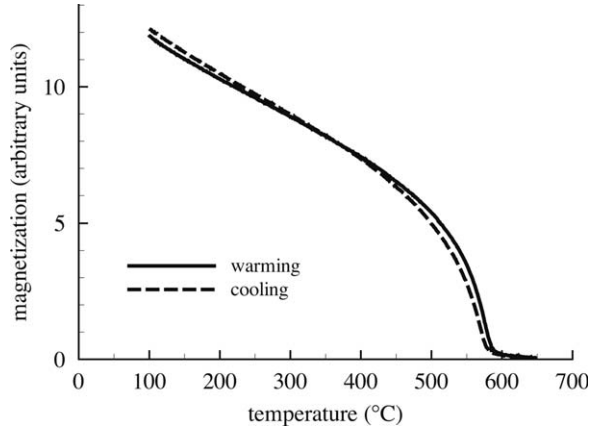


Figure 4. Example of thermomagnetic results. The Curie point, determined by the second-derivative method, is 579±4°C.

[6] A typical thermomagnetic result is shown in Figure 4. Heating and cooling curves are highly reproducible indicating that any thermally induced alteration is minimal. For the seven specimens studied, Curie temperatures, determined using the second-derivative method [*Tauxe*, 1998], all lie in the restricted range between 576°C and 582°C, suggesting near-stoichiometric magnetite. FORC measurements (Figure 5) reveal a low-coercivity peak at 18–20 mT followed by a high-coercivity tail extending beyond 200 mT.

3.2. Paleointensity Experiments

[7] A total of 25 mini-core specimens were used in paleointensity experiments using the double-heating Thellier-Thellier-Coe protocol with pTRM and pTRM-tail checks [*Coe*, 1967; *Walton*, 1984; *Riisager and Riisager*, 2001], using a laboratory field of 40 A/m (50.3 μT). The field was applied during both heating and cooling. The results were analyzed with the ThellierTool4.1 software of *Leonhardt et al*. [2004]. ThellierTool's default selection criteria (see Appendix 1) were used to classify the results, with one minor modification: due to the sharp unblocking spectra, the number of required points (*N)* was reduced from 5 to 4. Paleointensity estimates classed A and B were determined by maximizing the quality factor (*q*). Class C estimates were made by manually fitting lines to the Arai plots. Careful analysis yielded 12 A, 6 B, and 7 C results. An example from each category is shown in Figure 6, and the results for all 25 specimens are summarized in Table 1. Mean paleointensity values for categories A and B are 39±3 μT and 38±8 μT, respectively. Even the C category yields a similar value of 36±8 μT. The A and B results are characterized by very favorable "quality" criteria: median (and "worst case") values for *f, g, q* and MAD are 0.80 (0.36), 0.78 (0.51), 17 (3.6), and 1.5° (3.1°), respectively. For pTRM and pTRM tail checks, the corresponding values for δ (CK) and δ (TR) are 3.8 (8.1) and 2.1 (5.7), respectively.

4. Discussion

[8] The most important outcome of the work described here is the establishing of a robust paleointensity estimate of 38 μT, with a 95% confidence interval of 36–40 μT. The rock-magnetic characteristics of the material involved are close to ideal, leading to an unusually high success rate of 72% for the combined A and B classes. This is due to the fact that the stable remanence is carried by

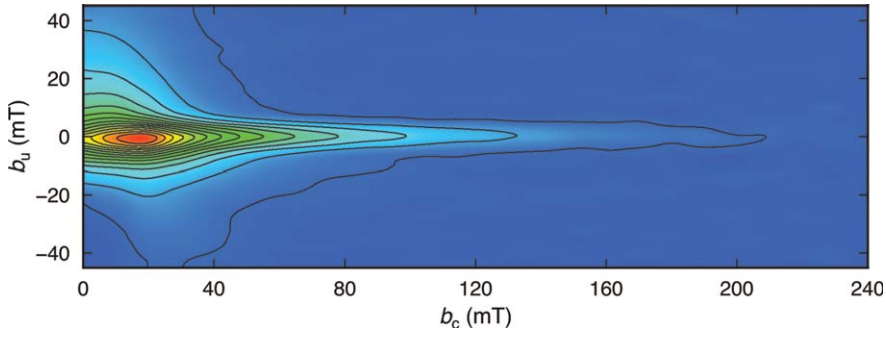


Figure 5. First-order reversal curve (FORC) diagram showing a peak at 18–20 mT, and a prominent high-coercivity tail. Smoothing factor is 3, and the averaging time 200 ms. One hundred and eighty FORCs were measured.

exsolved magnetite inclusions that are excellent paleomagnetic recorders protected by their silicate host from chemical change in nature and during laboratory experiments. Furthermore, this result is very well dated at 2784.0±1.5 Ma. The observed paleomagnetic directional data allows the paleointensity value to be converted to a virtual dipole moment (VDM) of $5.8\pm0.3\times10^{22}$ Am^2. The validity of this conversion is supported by evidence that the geomagnetic field was, indeed, dipolar in these remote times [*Smirnov et al*., 2011]. The calculated VDM itself could be halved if slow cooling were allowed for on the basis of SD theory [*Halgedahl et al*., 1980]. But cooling-rate corrections remain very much an open question, which we discuss in more detail further below. Another potential problem that may adversely affect paleointensity investigations is magnetic anisotropy. It can strongly influence results from rocks with a marked petrofabric, such as those studied by *Selkin et al*. [2000]. But such effects are unlikely to be significant in rocks like the Modipe Gabbro that have no noticeable petrofabric.

[9] Due to the inherent difficulties in extracting reliable data from very ancient rocks, only a handful of relevant paleointensity results are currently available. But some progress is being made [*Smirnov et al*., 2003; *Macouin et al*., 2004; *Tarduno et al*., 2007]. There are only three Thellier results with ages similar to the Modipe Gabbro (2.7–2.8 Ga): one from an anorthosite sample of the Stillwater Compex in Montana, USA, one from a lava flow in the Pilbara Craton, Australia, and one from a dolerite dyke in Greenland. For the Stillwater sample, *Selkin et al*. [2000] obtain a raw paleointensity of 92 μT, but after careful consideration of the magnetic anisotropy resulting from pronounced foliation, they modify this to 46 μT. They then apply a cooling-rate correction based on SD theory that further reduces the paleointensity to 32 μT, corresponding to a VDM of $4.1\pm0.5\times10^{22}$ Am^2. *Biggin et al*. [2009] undertook an exhaustive study of 54 samples from the Pilbara Craton, but were plagued by severe magnetomineralogical alteration during laboratory experiments. After application of a correction procedure to allow for such alteration, a single lava flow survived their strict acceptance criteria and yielded a VDM of $4.7\pm0.6\text{x}10^{22}$ Am^2. *Morimoto et al*. [1997] studied a dolerite dyke in west Greenland and found a VDM of $1.9\pm0.6\times10^{22}$ Am^2 (after rejecting a high outlier of 6.0×10^{22} Am^2). The Australian and Greenland results both come from geological formations that probably cooled rapidly, so the authors did not apply cooling corrections.

[10] An early attempt to obtain a paleointensity result from the Modipe Gabbro was made by *McElhinny and Evans* [1968], using samples from the same locality studied in the present paper (their site BO: 24°39'481S, 26°10'291E). Because of the restricted range of blocking temperatures, they opted to use a non-Thellier technique involving comparison of the NRM and TRM coercivity spectra. From the high-coercivity range 100–160 mT, they obtained a paleointensity estimate of 94±10 μT—well over twice the Thellier value we obtain. Conventional wisdom dictates that the Thellier result is to be preferred, and this is the position we adopt here. Reasons for seriously doubting the original AF procedure of *McElhinny and Evans* [1968] are given in a thorough investigation by *Kono* [1978].

4.1. A Preisach Approach to a Cooling Rate Correction

[11] Differences in cooling rates are known to strongly affect paleointensity estimates. It has been demonstrated experimentally that SD and MD grains respond in the opposite sense [*McClelland-Brown*, 1984; *Yu*, 2011]. Theoretically, *Halgedahl*

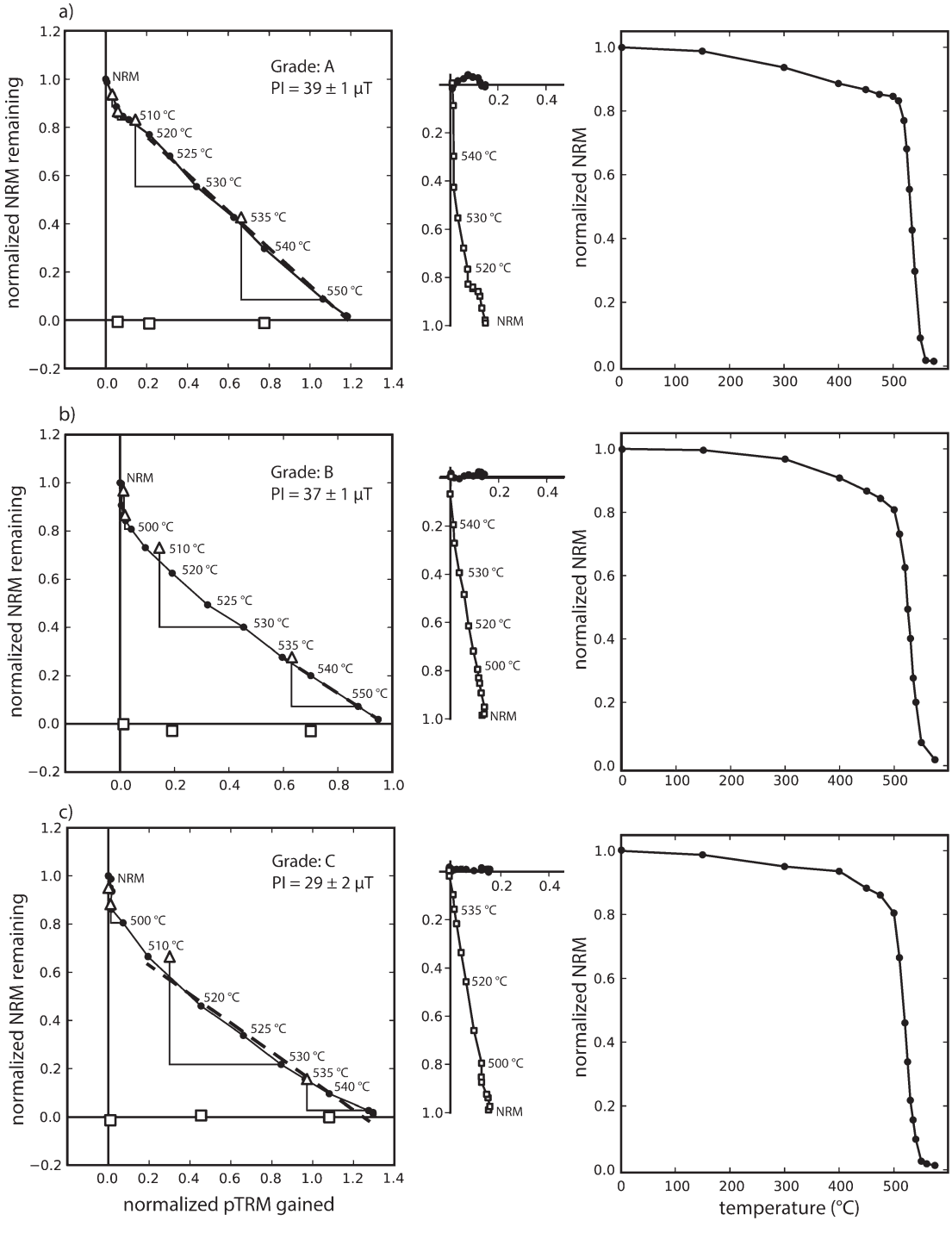


Figure 6. Examples of Arai plots for classes A, B, and C. Triangles are pTRM checks, and squares are pTRM-tail checks. Specimen identification corresponds to Table 1. Corresponding orthogonal projection plots (rotated) and NRM demagnetization plots derived from the palaeointensity experiment are also shown. (a) Specimen 24. (b) Specimen 6. (c) Specimen 2.

et al. [1980], argue that SD assemblages may require correcting downwards by as much as ∼50%, whereas *Winklhofer et al*. [1997] show that assemblages of small PSD particles may, in some cases, require even larger positive corrections; experimental work [*Yu*, 2011] indicates that corrections should be generally smaller than theory predicts. Generally, however, if a cooling-rate correction is made it is assumed that remanence is carried by SD grains. Our observed grain-size distribution for the Modipe Gabbro implies a significant PSD contribution [*McElhinny and Evans*,

Table 1. Thellier Results Measured for This Study[a]

Sample	Intensity (μT)	±σ (μT)	ΔT (°C)[b]	N	f	g	q	MAD (°)	δ (CK)	δ_{pal}	δ (TR)	Δ (t*)	Class
1	40	2	450–510	4	0.37	0.51	3.6	0.9	1.0	1.1	2.5	1.8	A
2	29	2	510–560	8	0.87	0.82	14	2	8.4	7.9	1.8	0	C
3	32	2	400–575	13	0.99	0.85	18	1.1	6.8	3.3	1.3	4.9	B
4	43	1	510–575	9	0.9	0.83	25	1.2	8.1	13.9	1.4	0.0	A
5	43	1	510–575	9	0.83	0.8	36	1.4	8.0	9.9	1.5	0.0	C
6	37	0.1	535–575	4	0.36	0.62	100	3.1	5.3	8.7	4.2	0.0	B
7	39	0.4	520–540	5	0.55	0.72	34	1.4	0.8	1	1.8	3.0	A
8	37	1	510–575	9	0.86	0.8	25	1.6	9.0	6.8	2.1	0.6	C
9	33	3	535–560	4	0.26	0.58	1.7	3.4	8.5	0.5	1.0	0.0	C
10	30	2	150–575	15	1.00	0.85	13	1.8	6.7	10	1.9	0.0	B
11	21	1	510–575	9	0.99	0.82	20	2.7	17	17.6	2.6	16.1	C
12	33	2	50–560	13	1.00	0.78	15	1.1	0.5	0.7	1.3	1.3	A
13	34	2	300–535	10	0.56	0.8	10	1.4	2.4	3.8	2.1	2.4	A
14	37	4	50–535	11	0.66	0.66	4.6	0.8	1.1	2	1.9	0.0	A
15	38	1	400–540	10	0.78	0.83	23	2.5	5.2	9.9	4.5	3.5	B
16	38	1	450 –575	10	0.94	0.77	34	1.9	4.0	4.1	1.7	0.0	A
17	39	1	535–575	5	0.40	0.66	12	2	3.1	4.4	2.2	2.6	A
18	44	1	510–575	7	0.91	0.74	25	1.6	9.9	7.4	2.6	0.0	C
19	40	6	150–575	15	0.93	0.75	4.8	2	6.2	7.5	5.7	4.9	B
20	52	4	525–540	4	0.44	0.66	4.0	1	3.5	8.1	3.8	4.9	B
21	42	2	510–560	7	0.87	0.79	13	1.4	2.7	1.8	4.2	5.5	C
22	43	3	520 –540	5	0.56	0.74	6.3	1.6	5.0	4.7	2.9	0.8	A
23	41	1	500–575	9	0.90	0.84	31	1.7	4.9	4.9	3.0	0.9	A
24	39	1	520–575	8	0.82	0.81	33	1.9	3.0	4.6	1.5	0.0	A
25	38	1	400–575	12	0.95	0.85	47	1	2.0	3.2	3.4	0.7	A

[a]Definitions of the various parameters are provided in Table A1.
[b]ΔT is the temperature range used to make the paleointensity estimate.

1968], so it seems unlikely that a blanket application of an SD-cooling correction is appropriate. Instead we use the Preisach model of *Muxworthy and Heslop* [2011], which has previously been successfully applied to the study of cooling-rate variation on TRM intensity [*Muxworthy et al.*, 2011; *Biggin et al.*, 2013]. The Preisach model describes the behavior of distributions of interacting uniaxial SD grains; however, we argue, that the inclusion of interactions makes this model a better approximation for PSD and MD behavior than the noninteracting SD models previously employed.

[12] The measured FORC distribution (Figure 5) is used as input to generate a corresponding Preisach distribution. During a Thellier experiment, different grain sizes and domain states become thermally blocked/unblocked at different

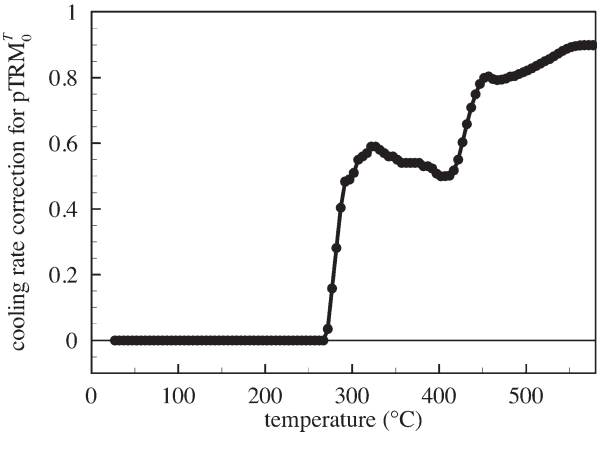


Figure 7. Cooling-rate correction as a function of temperature for a pTRM acquired between T and room-temperature (pTRM$_0^T$). The correction was calculated from the FORC diagram shown in Figure 5, assuming a cooling time of one million years.

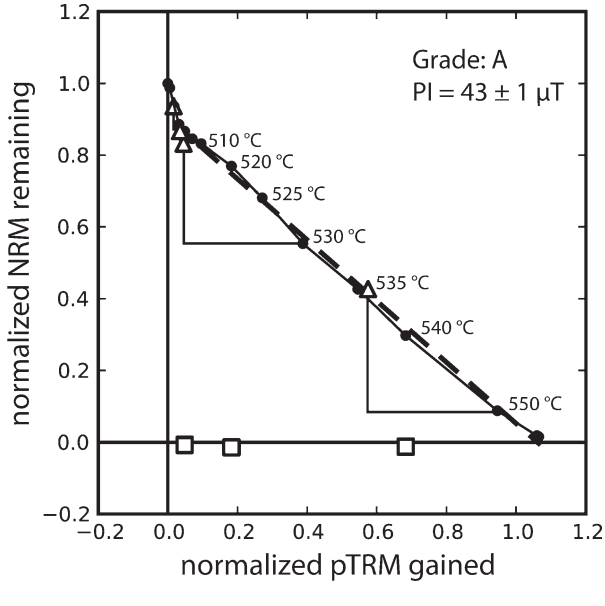


Figure 8. An example of cooling-corrected Arai plot for sample 24 (also shown in Figure 6a).

Table 2. Cooling-Rate-Corrected Thellier Results Measured for This Study[a]

Sample	intensity (μT)	±σ (μT)	ΔT (°C)[1]	N	f	g	q	δ (CK)	δ_{pal}	δ (TR)	Δ (t^*)	Class
1	45	3	450–510	4	0.37	0.51	2.9	0.5	1.8	2.6	1.9	A
2	32	2	510–575	9	0.89	0.82	13	5	4.6	1.9	0.0	A
3	50	4	50–530	11	0.83	0.78	7.6	4.6	7.6	1.2	5.0	B
4	52	2	475–575	11	1	0.85	21	6.5	9.6	1.3	0.0	B
5	45	2	520–575	8	0.6	0.79	14	4	3.4	1.6	0.0	A
6	46	2	510–575	8	0.92	0.85	24	3.5	4	3.9	0.0	A
7	50	2	450–560	10	1	0.82	17	0.9	0	1.6	3.0	A
8	41	1	510–575	5	0.54	0.72	25	5.4	0.5	2.2	0.7	B
9												C
10	18	1	525–540	4	0.45	0.63	7.9	3.8	3.3	3.4	0.0	A
11												C
12	37	2	50–560	13	1	0.78	14	0.7	0.6	1.3	1.4	A
13	38	2	475–535	7	0.5	0.75	6.8	3.8	9.4	2.1	2.7	B
14	40	5	50–575	16	1	0.49	3.8	4.3	2.4	2.2	0.0	B
15												C
16	42	1	450 –575	11	0.97	0.78	34	2.9	2.5	1.7	0.0	A
17	43	1	530 –575	6	0.57	0.75	17	4.5	4.9	2.2	3.0	B
18												C
19												C
20	43	4	530–540	6	0.6	0.75	4.5	4.2	8.9	4.4	6.3	B
21	43	1	50–575	15	0.98	0.86	35	4.3	8.8	3	0.9	B
22												C
23												C
24	43	1	400–575	11	0.97	0.85	60	4.7	2.5	1.6	0.0	A
25	42	1	400–575	11	0.97	0.86	40	5.3	5.5	3.5	0.8	B

[a]Definitions of the various parameters are provided in Table A1.

temperatures. The cooling-rate correction will therefore vary as a function of temperature during the experiment. We use the Preisach model to determine this temperature-dependent cooling rate correction. Assuming an original cooling time of a million years, a cooling-rate correction curve as a function of temperature is shown in Figure 7. Using this curve, we individually correct for cooling rate at each point in the Arai plots (e.g., Figure 8), and retabulate the paleointensity estimates (Table 2). The mean paleointensity value for the combined categories A and B is now 42±6 μT (95% confidence interval 39–45 μT), a modest *increase* of ∼10% from the uncorrected estimate. As a first approximation the correction is linear with cooling time, e.g., 0.5 million years gives an increase of ∼5%.

5. Conclusions

[13] Thellier-Thellier-Coe paleointensity results are reported from a new collection of samples from the Modipe Gabbro of Botswana. Very favorable rock-magnetic properties lead to an unusually high success rate, and the result obtained corresponds to a VDM of $5.8\pm0.3\times10^{22}$ Am^2 at 2.78 Ga. After a temperature-dependent cooling-rate correction is applied this is increased to $6.4\pm0.4\times10^{22}$ Am^2. Thus, a significant geomagnetic field was certainly present during the late Archaean, with a strength up to 80% of the modern value.

Appendix

Table A1. Summarizing the Thellier Tool 4.1 Default Criteria[a]

Criteria Description	Class A	Class B
Linear fit criteria		
Number of points (*N*) used to determine the paleointensity	≥5	≥5
Normalized standard deviation of slope (β)	≤0.1	≤0.15
Fraction of NRM (*f*)	≥0.3	≥0.3
Quality factor (*q*)	≥1	≥0
Directional criteria		
Maximum angular deviation (MAD) of the anchored fit	≤6°	≤15?
Angular difference between the anchored and nonanchored solution (α)	≤15°	≤15?
Alteration criteria		
Maximum difference produced by a pTRM check, normalized by the TRM (δ (CK))	≤5 %	≤7 %
Cumulative difference produced by pTRM checks (δ_{pal})	≤5 %	≤10 %
Repeated demagnetization steps		
Extent of pTRM tail after correction for angular dependence pTRM (δ (t^*))	≤3 %	≤5 %
Maximum difference produced by a pTRM tail check, normalized by the NRM (δ (TR))	≤10 %	≤20 %

[a]These parameters are described in more detail in *Leonhardt et al.* [2004] and references therein. Common abbreviations for the symbols are bracketed.

References

Aubert, J., J. A. Tarduno, and C. L. Johnson (2010), Observations and models of the long-term evolution of earth's magnetic field, *Space Sci. Rev.,* doi:10.1007/s11214-010-9684-5.

Biggin, A., G. H. M. A. Strik, and C. G. Langereis (2009), The intensity of the geomagnetic field in the late-Archaean: New measurements and an analysis of the updated IAGA palaeointensity database, *Earth Planets Space*, *61*, 9–22.

Biggin, A. J., S. Badejo, E. Hodgson, A. R. Muxworthy, J. Shaw, and M. J. Dekkers (2013), The effect of cooling rate on the intensity of thermoremanent magnetisation (TRM) acquired by assemblages of pseudo-single domain, multi domain, and interacting single domain grains, *Geophys. J. Int.*, in press.

Coe, R. S. (1967), The determination of paleointensities of the Earth's magnetic field with emphasis on mechanisms which could cause non-ideal behavior in Thellier's method, *J. Geomagn. Geoelectr.*, *19*, 157–179.

Day, R., M. D. Fuller, and V. A. Schmidt (1977), Hysteresis properties of titanomagnetites: Grain-size and compositional dependence, *Phys. Earth Planet. Inter.*, *13*, 260–267.

Evans, M. E., and M. W. McElhinny (1966), The paleomagnetism of the Modipe gabbro, *J. Geophys. Res.*, *71*(24), 6053–6063.

Evans, M. E., M. W. McElhinny, and A. C. Gifford (1968), Single domain magnetite and high coercivities in a gabbroic intrusion, *Earth Planet. Sci. Lett.*, *4*, 142–146.

Feinberg, J. M., S. W. Denyszyn, P. R. Renne, and G. R. Scott (2010), The paleomagnetism and age of the Modipe Gabbro, South Africa, *Geochim. Cosmochim. Acta*, *74*(12), A283–A283.

Halgedahl, S., R. Day, M. Fuller (1980), The effect of cooling rate on the intensity of weak-field TRM in single-domain magnetite, *J. Geophys. Res.*, *85*, 3690–3698.

Kono, M. (1978), Reliability of palaeointensity methods using alternating field demagnetization and anhysteretic remanence, *Geophys. J. R. Astr. Soc.*, *54*, 241–261.

Leonhardt, R., C. Heunemann, and D. Krása (2004), Analyzing absolute paleointensity determinations: Acceptance criteria and the software ThellierTool4, *Geochem. Geophys. Geosyst.*, *5*, Q12016, doi:10.1029/2004GC000807.

Macouin, M., J. P. Valet, and J. Besse (2004), Long-term evolution of the geomagnetic dipole moment, *Phys. Earth Planet. Inter.*, *147*, 239–246, doi:10.1016/j.pepi.2004.07.003.

McClelland-Brown, E. (1984), Experiments on TRM intensity dependence on cooling rate, *Geophys. Res. Lett.*, *11*, 205–208.

McElhinny, M. W., and M. E. Evans (1968), An investigation of the strength of the geomagnetic field in the early Precambrian, *Phys. Earth Planet. Inter.*, *1*, 485–497.

Morimoto, C., Y. Otofuji, M. Miki, H. Tanaka, and T. Itaya (1997), Preliminary palaeomagnetic results of an Archaean dolerite dyke of west Greenland: Geomagnetic field intensity at 2.8 Ga, *Geophys. J. Int.*, *128*, 585–593.

Muxworthy, A. R. and D. Heslop (2011), A Preisach method for estimating absolute paleofield intensity under the constraint of using only isothermal measurements: 1. Theoretical framework, *J. Geophys. Res.*, *116*, B04102, doi:10.1029/2010JB007843.

Muxworthy, A. R., and M. E. Evans (2013), Micromagnetics and magnetomineralogy of Ultrafine magnetite inclusions in the Modipe Gabbro, *Geochem. Geophys. Geosystems*, in press.

Muxworthy, A. R., D. Heslop, G. A. Paterson, and D. Michalk (2011), A Preisach method for estimating absolute paleofield intensity under the constraint of using only isothermal measurements: 2. Experimental testing, *J. Geophys. Res.*, *116*, B04103, doi:10.1029/2010JB007844.

Riisager, P., and J. Riisager (2001), Detecting multidomain magnetic grains in Thellier paleointensity experiments, *Phys. Earth Planet. Inter.*, *125*(1), 111–117.

Selkin, P. A, J. S. Gee, L. Tauxe, W. P. Meurer, and A. J. Newell (2000), The effect of remanence anisotropy on paleointensity estimates: A case study from the Archean Stillwater Complex, *Earth Planet. Sci. Lett.*, *183*, 403–416.

Smirnov, A. V., J. A. Tarduno, and B. N. Pisakin (2003), Paleointensity of the early geodynamo (2.45 Ga) as recorded in Karelia: A single-crystal approach, *Geology*, *31*, 415–418.

Smirnov, A. V., J. A. Tarduno, and D. A. D. Evans (2011), Evolving core conditions ca. 2 billion years ago detected by paleosecular variation, *Phys. Earth Planet. Inter.*, *187*, 225–231.

Tarduno, J. A. (2009), Geodynamo history preserved in single silicate crystals: Origins and long-term mantle control, *Elements*, *5*(4), 217–222.

Tarduno, J. A, R. D. Cottrell, M. K. Watkeys, and D. Bauch (2007), Geomagnetic field strength 3.2 billion years ago recorded by single silicate crystals, *Nature*, *446*, 657–660. doi:10.1038/nature05667.

Tarduno, J. A., R. D. Cottrell, M. K. Watkeys, A. Hofmann, P. V. Doubrovine, E. E. Mamajek, D. Liu, D. G. Sibeck, L. P. Neukirch, and Y. Usui (2010), Geodynamo, solar wind, and magnetopause 3.4 to 3.45 billion years ago, *Science*, *327*, 1238–1240.

Tauxe, L. (1998), *Paleomagnetic Principles and Practice*, 299 pp., Kluwer Acad., Dordrecht, Netherlands.

Walton, D. (1984), Re-evaluation of Greek archaeomagnetic intensities, *Nature*, *310*, 740–743.

Winklhofer, M., K. Fabian, and F. Heider (1997), Magnetic blocking temperatures of magnetite calculated with a three-dimensional micromagnetic model, *J. Geophys. Res.*, *102*, 22,695–22,709.

Yu, Y. (2011), Importance of cooling rate dependence of thermoremanence in paleointensity determination, *J. Geophys. Res.*, *116*, B09101, doi:10.1029/2011JB008388.